Predictive biomarkers for sustained unresponsiveness following oral immunotherapy

Duan Ni[1,2,3,*], Ralph Nanan[1,2,3,*]

1 Sydney Medical School Nepean, The University of Sydney, Sydney, NSW, Australia

2 Charles Perkins Centre, The University of Sydney, Sydney, NSW, Australia

3 Nepean Hospital, Nepean Blue Mountains Local Health District, Sydney, NSW, Australia

*Correspondence:

Ralph Nanan

ralph.nanan@sydney.edu.au

Duan Ni

duan.ni@sydney.edu.au

Sydney Medical School Nepean, The University of Sydney

Nepean Hospital, Level 5, South Block, Penrith NSW, 2751, Australia

Telephone: +61 2 4734 1614

Fax: +61 2 4734 1144

Table 1. Overview of the state-of-the-art advances in predicting sustained unresponsiveness (SU) outcome from oral immunotherapy (OIT) with baseline measurements and receiver operating characteristic (ROC) analyses for 3 studies with publicly available source data and the *Oral Peanut Immunotherapy with butyrate Adjuvant* (OPIA, ACTRN12617000914369) clinical trial. Data were curated from 22 papers based on 11 clinical trials/studies.

(POISED trial: Peanut Oral Immunotherapy: Safety, Efficacy, Discovery trial; ML: machine learning; esIgE: epitope-specific IgE; esIgG4: epitope-specific IgG4; sIgE: specific IgE; ACU: area under the curve; CRD: cumulative dose; GzB: granzyme B; EM: effector memory; EMRA: effector memory re-expressing CD45RA; CyTOF: cytometry by time of flight; mDC2: type 2 myeloid dendritic cell; NK cell: natural killer cell; MFI: mean fluorescence intensity; pr: peanut reactive; scRNA-seq: single cell RNA sequencing; UMM: untargeted metabolomics module; BA: bile acid; ME: module eigenvector; AA: amino acid; SBA: secondary bile acid; HAMSB: butyrylated high amylose maize starch; LAMS: low amylose maize starch; n.s.: not significant; q: q values were computed using the $\chi 2$ tests in mixed-effects models, adjusted for multiple hypothesis testing using the Benjamini and Hochberg method)

Overview of advances in predicting sustained unresponsiveness (SU) outcome from oral immunotherapy (OIT) with baseline measurements.							
Clinical trial ID	**Allergen**	**Study design**	**Baseline measurement**	**Statistical output**	**# participants**	**Methods**	**References**
NCT02103270	Peanut	POISED trial PN-0: 2y OIT, 1y avoidance (SU outcome) PN-300: 2y OIT, 1y 300mg daily peanut (SHT (sustained high threshold) outcome) (PN-0, PN-300 combined for analysis)	ML features based on esIgE & peanut sIgE	AUC=0.99	21 SU/SHT vs 56 no SU/SHT	Luminex-based peptide assay	[1]
			ML features based on esIgE & CRD	AUC=0.99			
			ML features based on esIgE & age	AUC=0.97			
			ML features based on esIgE & esIgG4	AUC=0.97			
			ML features based on esIgE	AUC=0.97			
			CRD	AUC=0.74		Clinical measurements	
		POISED trial PN-0: 2y OIT, 13w avoidance timepoint	Peanut-stimulated %Gnly+GzB- of CD8+ T	SU > no SU, p=0.0368	8 SU, 7 no SU	Spectral flow cytometry	[2]
			Peanut-stimulated %Gnly-GzB+ of CD8+ T	SU > no SU, p=0.0127			
		POISED trial, PN-0	HLA-DQA1*01:02 carry status.	SU (12/20), no SU (11/29), n.s.	20 SU, 29 no SU	Whole genome sequencing	[3]
		POISED trial, PN-300		SHT (4/18), no SHT (1/10), n.s.			

		POISED trial, PN-0	Basophil assay AUC in response to peanut extract protein	SU < no SU, p=0.04	20 SU, 22 no SU	Basophil activation assay	
			Peanut sIgE	SU < no SU, p<0.0001		ImmunoCAP	
			Ara h 1 sIgE	SU < no SU, p<0.0001			
			Ara h 2 sIgE	SU < no SU, p<0.0001			
			Ara h 3 sIgE	SU < no SU, p=0.002			
			Peanut sIgE/total IgE	SU < no SU, p=0.002			
			Peanut sIgG4	SU < no SU, p=0.0006			
			Peanut sIgG4/sIgE	SU > no SU, p=0.01			
		POISED trial, PN-300	Basophil assay AUC in response to peanut extract protein	SHT < no SHT, p=0.06	18 SHT, 10 no SHT	Basophil activation assay	[4]
			Peanut sIgE	SHT < no SHT, p=0.05		ImmunoCAP	
			Ara h 1 sIgE	SHT < no SHT, p=0.03			
			Ara h 2 sIgE	SHT < no SHT, p=0.02			
			Ara h 3 sIgE	SHT < no SHT, p=0.06			
			Peanut sIgE/total IgE	SHT < no SHT, p=0.04			
		POISED trial PN-0: 2y OIT, 13w avoidance timepoint	Peanut-stimulated %CD57+ of EM CD8+ T	SU < no SU, q=0.019	20 SU, 29 no SU	CyTOF	[5]
			Peanut-stimulated %CD57+ of EMRA CD8+ T	SU < no SU, q=0.023	20 SU, 29 no SU		
			Peanut-stimulated %naive CD8+ T of total live cell	SU < no SU, q=0.024	20 SU, 29 no SU		
			Unstimulated %CD57+ of EM CD8+ T	SU < no SU, q=0.049	20 SU, 26 no SU		

			Unstimulated %CD57+ of EMRA CD8+ T	SU < no SU, q=0.034	20 SU, 26 no SU		
			Unstimulated %naive CD8+ T of total live cell	SU < no SU, q=0.025	20 SU, 26 no SU		
			Peanut-stimulated %CD86+ of memory B	SU < no SU, q=0.038	20 SU, 29 no SU		
			Unstimulated %CD86+ of classical monocyte	SU < no SU, q=0.053	20 SU, 26 no SU		
			Unstimulated %CD86+ of mDC2	SU < no SU, q=0.030	20 SU, 26 no SU		
			Peanut-stimulated %CD57+ of CD56dimCD16+ NK cell	SU < no SU, q=0.06	20 SU, 29 no SU		
			Unstimulated %CD57+ of CD56dimCD16+ NK cell	SU < no SU, q=0.03	20 SU, 26 no SU		
			Peanut-stimulated PBMC culture supernatant IL2 MFI	SU < no SU, q=0.056	7 SU, 7 no SU		
			ML features based on cell frequencies and marker expression of all subsets	AUC=0.94	20 SU, 29 no SU		
			Th2 cytokine module score of peanut-stimulated CD4+CD154+ pr effector T cell	AUC=0.7959	7 SU, 7 no SU		
			Asthma pathway gene module score of peanut-stimulated CD4+CD154+ pr effector T cell	AUC=0.8163	7 SU, 7 no SU	scRNA-seq	[6]
ACTRN12616000322437	Peanut	PPOIT-003 trial 18m placebo/OIT/probiotic and OIT (PPOIT) 2m avoidance (only OIT, PPOIT arms combined for analysis)	Peanut sIgE	AUC=0.89 (95% CI: 0.84-0.95)	78 SU, 83 no SU	ImmunoCAP	[7]
			Age	AUC=0.64 (0.55-0.74)			
			Age <6	SU (51/80) > no SU (53/121), p=0.006	80 SU, 121 no SU	Clinical measurements	[8]
			Skin prick test wheal size <=10mm	SU (43/80) > no SU (36/121), p=0.001			
			Have >=2 food allergy?	SU (28/80) < no SU (68/121), p=0.004			
			Have wheeze?	SU (39/80) < no SU (78/121), p=0.028			

			Have anaphylaxis?	SU (25/80) < no SU (55/121), p=0.045			
			Have asthma?	SU (17/80) < no SU (46/121), p=0.013			
			Peanut eliciting dose >320mg?	SU (48/80) > no SU (39/121), p<0.001			
			Peanut sIgE >=40kU/L?	SU (5/77) < no SU (61/124), p<0.001			
ACTRN 12608000594325	Peanut	PPOIT trial 18m OIT, 4y post treatment	Peanut sIgE	SU 1.29 (0.26, 6.32) < no SU 85.34 (16.73, 435.41), (geometric mean, 95%CI), p=0.003	7 SU, 5 no SU	ImmunoCAP	[9]
			Ara h 1 sIgE	SU 0.11 (0.00, 3.29) < no SU 27.20 (3.49, 212.1), p=0.012			
			Ara h 2 sIgE	SU 0.67 (0.15, 3.10) < no SU 38.91 (12.01, 126.02), p=0.001			
			Ara h 3 sIgE	SU 0.03 (0.00, 0.16) < no SU 1.48 (0.03, 84.28), p=0.044			
			Peanut sIgG4	SU 0.21 (0.11, 0.40) < no SU 0.62 (0.44, 0.87), p=0.025			
			Ara h 1 sIgG4	SU 0.01 (0.01, 0.03) < no SU 0.13 (0.02, 0.94), p=0.063			
			Ara h 2 sIgG4	SU 0.07 (0.05, 0.09) < no SU 0.18			

				(0.06, 0.54), p=0.055			
NCT01324401	Peanut	56w OIT, 4w avoidance	Ara h 2 sIgE	SU (8.6 ± 1.7 kU/L) < no SU (70.3 ± 1.6 kU/L), p=0.002,	9 SU, 13 no SU	ImmunoCAP	[10]
			Ara h 2 sIgE/Ara h 2 sIgG	SU (3.0 ± 3.4) < no SU (27.4 ± 3.1), p=0.003,			
			Ara h 2 sIgE/Ara h 2 sIgG4	SU (21.4 ± 4.0) < no SU (135.8 ± 3.6), p=0.006			
			Ara h 2 sIgE/total IgE	n.s.			
			Age of diagnosis	SU (1.44) < no SU (2.23), p=0.02		Clinical measurements	
			Basophil assay AUC in response to Ara h 2	n.s.		Basophil activation assay	
			Basophil sensitivity	n.s.			
NCT01867671	Peanut	IMPACT trial 134w OIT, 26w avoidance	Peanut sIgE	SU < DS & Fail, p<0.01	19 SU, 40 DS (no SU but desensitized), 10 Fail (no SU, no DS)	ImmunoCAP	[11]
			Peanut sIgE/total IgE	SU < DS & Fail, p<0.01			
			Peanut sIgG4	n.s.			
			Peanut sIgE/peanut sIgG4	SU < DS & Fail, p<0.05			
			Ara h 2 sIgE	SU < DS & Fail, p<0.01			
			Skin prick test wheal size	n.s.		Clinical measurements	
			Basophil assay % $CD63^+$ basophil	SU < DS & Fail, p<0.01		Basophil activation assay	
			Peanut sIgG4 in saliva	n.s.		ELISA	[12]
			Total IgG4 in saliva	n.s.			
			Peanut sIgG4/total IgG4 in saliva	n.s.			
			Peanut sIgA in saliva	SU < Fail, p<0.05			
			Total IgA in saliva	SU < Fail, p<0.05			
			Peanut sIgA/total IgA in saliva	n.s.			

			Th2A immunotype	SU < DS & Fail No stats provided	7 SU, 8 DS, 5 Fail	Flow cytometry	[13]
			Age at screening (month)	SU (31 (IQR, 24-40)), DS (39 (33-45)), Fail (40 (39-42)), p=0.01	20 SU, 49 DS, 12 Fail	Clinical measurements	[14]
			Cumulative tolerated dose of masked DBPCFC to peanut at baseline (mg)	SU (75 (25-300)), DS (25 (5-75)), Fail (50 (20-75)), p=0.045			
			Ara h 2 sIgE	SU (13 (8-39)), DS (64 (35-87)), Fail (107 (56-163)), p<0.01		ImmunoCAP	
			Peanut sIgE	SU (18 (12-50)), DS (67 (39-195)), Fail (243 (62-359)), p<0.01			
			Peanut sIgE/total IgE	SU (8 (3-19)), DS (24 (15-43)), Fail (33 (22-61)), p<0.01			
			Peanut sIgG4/peanut sIgE	SU (0.006 (0.002-0.047)), DS (0.003 (0.001-0.010)), Fail (0.002 (0.001-0.005)), p=0.03			
			Gut microbiota phylogenetic diversity	SU vs DS, p=0.01; SU vs Fail, p=0.05	16 SU, 23 DS (no SU but desensitized), 8 Fail (no SU, no DS)	Gut microbiota metagenomic analysis	[15]
			UMM 10 secondary BA metabolism ME	SU vs DS, p=0.0071; SU vs Fail, p=0.065		Untargeted metabolomics	
			UMM15 Secondary BA metabolism ME	SU vs DS, p=0.03; SU vs Fail, p=5.8e-05			
			UMM4 AAs, SBAs, xenobiotics ME	SU vs DS, p=0.019; SU vs Fail, p=0.0023			

			ML logistic regression model based on metabolites, 3-dehydrocholate, chenodeoxycholate, catechol sulfate, 7-ketolithocholate, 7-ketodeoxycholate	AUC=0.712 ± 0.081			
N/A	Peanut	38.5m OIT, 4w avoidance	Skin prick test wheal size	SU (median, 9mm) < DS (14mm), p=0.02	12 SU, 12 no SU	Clinical measurements	[16]
			Peanut sIgE	SU < no SU, p<0.01		ImmunoCAP	
			Ara h 1 sIgE	SU < no SU, p<0.01			
			Ara h 2 sIgE	SU < no SU, p<0.05			
			Peanut sIgE/total IgE	SU < no SU, p<0.01			
			Peanut sIgG4	n.s.			
			Peanut sIgE/peanut sIgG4	SU < no SU, p<0.05			
NCT00461097	Egg	20m-4y OIT, 4-6w avoidance	OVM sIgE	SU (3.97 kU/l) < no SU (10.9 kU/l), p=0.011, AUC=0.74 (0.58, 0.91 95% CI)	19 SU, 18 no SU	ImmunoCAP	[17]
			Egg white sIgE	SU (9.59 kU/l) < no SU (20.03 kU/l), p=0.052, AUC=0.69 (0.51, 0.86)			
			OVM sIgG4/OVM sIgE	AUC=0.70 (0.51, 0.89)			
			Egg sIgE	SU (9.4kUA/l) < no SU (18.5kUA/l), p=0.07	20 SU, 20 no SU		[18]
			Egg sIgE/total IgE	SU (1.1%) < no SU (2.7%), p=0.04			
			Log10 egg sIgE	AUC=0.67 (0.50-0.84)			
	Egg	PREMO study	Skin prick test wheal size <=10mm?	n.s.	11 SU, 9 no SU		[19]

ACTRN12619000306112	Milk	18m OIT & probiotics Lactobacillus rhamnosus GG (ATCC 53103), 8w avoidance	Skin prick test wheal size <=10mm?	n.s.	10 SU, 10 no SU	Clinical measurements	
IRCT20190612043872N1	Wheat	14-17m OIT, 3-4m avoidance	Skin prick test wheal size	n.s.	13 SU, 6 no SU	Clinical measurements	[20]
			Wheat sIgE	SU (106.0 kU/L) < no SU (161.7 kU/L), p=0.0008		ImmunoCAP	
			Wheat sIgG4	n.s.			
			Wheat sIgG4/wheat sIgE	n.s.			
NCT01157117	Milk	2y OIT +/- omalizumab, another 2m of OIT, 8w avoidance (+/- omalizumab arms combined for analysis)	Overall levels of esIgE	SU < no SU, p=0.001	23 SU, 21 no SU	Luminex-based peptide assay	[21]
			esIgE diversity	SU (76%) < DS (92%), p<0.01			
			Milk sIgE	SU (23.7 kUa/L, (5.1-38.9)) (mean, 1st & 3rd quartiles) < DS (55.6 kUa/L, (25.8-137.0)), p=0.01			
			Casein sIgE	SU (14.6 (1.9-38.3)) < DS 68.1 (23.0-163.0), p=0.003			
			β-Lactoglobulin sIgE	SU (3.2 (1.4-13.8)) < DS 15.9 (3.0-40.3), p=0.02			
			ML model with the 6 most informative IgE epitope results	AUC=0.95			
			Skin prick test wheal size	SU 7.5±2.4 < DS 10.2±3.4, p=0.004		Clinical measurements	
NCT05590299	Cod fish	1y OIT, 8w avoidance (OIT and placebo arms combined for analysis)	Codfish sIgE	n.s.	11 SU, 8 no SU	ImmunoCAP	[22]
			Codfish sIgG4/codfish s IgE	n.s.			
			rGad c1 sIgE	n.s.			
			Codfish sIgE/total IgE	SU < no SU, p=0.03			

			rGad c1 sIgE/total IgE	SU < no SU, p=0.042			
			Skin prick test wheal size	n.s.		Clinical measurements	
ROC analyses for 3 studies with publicly available source data							
Clinical trial ID	**Allergen**	**Study design**	**Baseline measurement**	**Statistical output**	**# participants**	**Methods**	**PMID**
IRCT20190612043872N1	Wheat	14-17m OIT, 3-4m avoidance	Wheat sIgE	AUC=0.9231 (0.799, 1) (95% CI)	13 SU, 6 no SU	ImmunoCAP	[20]
NCT01867671	Peanut	IMPACT trial 134w OIT, 26w avoidance	Gut microbiota phylogenetic diversity	AUC=0.7419 (0.5586, 0.9253)	16 SU, 31 no SU	Gut microbiota metagenomic analysis	[15]
			Peanut sIgE	AUC=0.8558 (0.7405, 0.9712)		ImmunoCAP	
			Ara h 2 sIgE	AUC=0.8344 (0.7069, 0.9620)			
NCT02103270	Peanut	POISED trial PN-0: 2y OIT, 13w avoidance timepoint	Peanut-stimulated %CD57+ of EM CD8+ T	AUC=0.6914 (0.5346, 0.8482)	20 SU, 29 no SU	CyTOF	[5]
			Peanut-stimulated %CD57+ of EMRA CD8+ T	AUC=0.5957 (0.4345, 0.7569)	20 SU, 29 no SU		
			Peanut-stimulated %naive CD8+ T of total live cell	AUC=0.7302 (0.5885, 0.8718)	20 SU, 29 no SU		
			Unstimulated %CD57+ of EM CD8+ T	AUC=0.6654 (0.5034, 0.8274)	20 SU, 26 no SU		
			Unstimulated %CD57+ of EMRA CD8+ T	AUC=0.5913 (0.4234, 0.7593)	20 SU, 26 no SU		
			Unstimulated %naive CD8+ T of total live cell	AUC=0.7240 (0.5770, 0.8710)	20 SU, 26 no SU		
			Peanut-stimulated %CD86+ of memory B	AUC=0.6784 (0.5159, 0.8410)	20 SU, 29 no SU		
			Unstimulated %CD86+ of classical monocyte	AUC=0.6837 (0.5270, 0.8403)	20 SU, 26 no SU		
			Unstimulated %CD86+ of mDC2	AUC=0.6010 (0.4340, 0.7679)	20 SU, 26 no SU		
			Peanut-stimulated %CD57+ of CD56dimCD16+ NK cell	AUC=0.6388 (0.4719, 0.8057)	20 SU, 29 no SU		
			Unstimulated %CD57+ of CD56dimCD16+ NK cell	AUC=0.6481 (0.4795, 0.8167)	20 SU, 26 no SU		

			Peanut sIgE	AUC=0.7986 (0.6451, 0.9552)	20 SU, 29 no SU		
			Ara h 2 sIgE	AUC=0.8287 (0.6789, 0.9785)	20 SU, 29 no SU		
			Ara h 2 sIgG4	AUC=0.5567 (0.3613, 0.7521)	20 SU, 29 no SU		
			Ara h 2 sIgE/Ara h 2 sIgG4	AUC=0.9190 (0.8141, 1.000)	20 SU, 29 no SU	ImmunoCAP	
ROC analyses for the in-house *Oral Peanut Immunotherapy with butyrate Adjuvant* (OPIA, ACTRN12617000914369) clinical trial							
ACTRN12617000914369	Peanut	OPIA trial 1y OIT supplemented with HAMSB/LAMS, 6w avoidance (HAMSB, LAMS arms combined for analysis)	Peanut eliciting dose	AUC=0.7315 (0.5940, 0.8689)	20 SU, 27 no SU	Clinical measurements	[23, 24]
			Peanut cumulative reactive dose	AUC=0.7370 (0.5992, 0.8749)			
			Peanut-stimulated %IFNγ+ of CD4+CD154+ pr effector T cell	AUC=0.6481 (0.4913, 0.8049)		Spectral flow cytometry	
			Peanut-stimulated %IL-4+ of CD4+CD154+ pr effector T cell	AUC=0.6722 (0.5088, 0.8356)			
			Peanut-stimulated %IL-5+ of CD4+CD154+ pr effector T cell	AUC=0.6806 (0.5212, 0.8399)			
			Peanut-stimulated %IL-10+ of CD4+CD154+ pr effector T cell	AUC=0.6046 (0.4362, 0.7730)			
			Peanut-stimulated %IL-13+ of CD4+CD154+ pr effector T cell	AUC=0.7000 (0.5449, 0.8551)			

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